AI Assisted Coding (III Year) Assignment

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Lab 16 – Database Design and Queries (AI Assisted)

Lab Objectives

- **Use AI to assist in designing and implementing fundamental data structures in Python**
- **Learn prompt-based structure creation, optimization, and documentation**
- **Improve understanding of Lists, Stacks, Queues, Linked Lists, Graphs, and Hash Tables**
- **Enhance readability and performance with AI-generated suggestions**

Task 1 – Student Information System Schema

1. AI Prompt Used

```
Generate SQL schema for a Student Information System.

Tables required:
Students
Courses
Enrollments

Requirements:

• Define primary keys
• Define foreign key relationships
• Maintain referential integrity
• Generate SQL queries for:
  Insert a new student
  Fetch all courses enrolled by a student
  Count students in each course
```

CODE and OUTPUT:

```
%%sql
INSERT INTO Courses (course_id, course_name)
VALUES (101, 'Mathematics');

 * sqlite:///lab16.db
1 rows affected.
[]
```

```
%%sql
INSERT INTO Enrollments (student_id, course_id)
VALUES (1, 101);

 * sqlite:///lab16.db
1 rows affected.
[]
```

```
%%sql
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY,
    name TEXT NOT NULL,
    email TEXT UNIQUE
);

CREATE TABLE Courses (
    course_id INTEGER PRIMARY KEY,
    course_name TEXT NOT NULL
);

CREATE TABLE Enrollments (
    enrollment_id INTEGER PRIMARY KEY,
    student_id INTEGER,
    course_id INTEGER,
    FOREIGN KEY(student_id) REFERENCES Students(student_id),
    FOREIGN KEY(course_id) REFERENCES Courses(course_id)
);

 * sqlite:///lab16.db
Done.
Done.
Done.
[]
```

```
%%sql
INSERT INTO Students (student_id, name, email)
VALUES (1, 'Akshay', 'akshay@email.com');

 * sqlite:///lab16.db
1 rows affected.
[]
```

```
import pandas as pd
df = %%sql --dataframe SELECT Students.name, Courses.course_name FROM Enrollments JOIN Students ON Students.student_id = Enrollments.student_id JOIN Courses ON Courses.course_id = Enrollments.course_id WHERE Students.student_id = 1;
display(df)

 * sqlite:///lab16.db
None
```

```
import pandas as pd
df = %%sql --dataframe SELECT Courses.course_name, COUNT(Enrollments.student_id) AS total_students FROM Enrollments JOIN Courses ON Courses.course_id = Enrollments.course_id GROUP BY Courses.course_name;
display(df)

 * sqlite:///lab16.db
None
```

Students

student_id	name	email
1	Akshay	akshay@email.com
2	Ravi	ravi@email.com

Courses

course_id	course_name
101	Database Systems
102	Machine Learning

Enrollments

enrollment_id	student_id	course_id
1	1	101
2	1	102
3	2	101

Task 2 – Hospital Management Database

AI Prompt

Generate SQL schema for a Hospital Management System.

Tables:

Doctors

Patients

Appointments

Requirements:

- Primary keys
- Foreign keys
- Date validation
- Queries for:

  List appointments for a doctor

  Retrieve patient history

  Count patients treated by each doctor

Schema AND Output:

```
CREATE TABLE Patients (
    patient_id INTEGER PRIMARY KEY,
    name TEXT,
    age INTEGER
);

CREATE TABLE Appointments (
    appointment_id INTEGER PRIMARY KEY,
    doctor_id INTEGER,
    patient_id INTEGER,
    appointment_date DATE,
    FOREIGN KEY(doctor_id) REFERENCES Doctors(doctor_id),
    FOREIGN KEY(patient_id) REFERENCES Patients(patient_id)
);

 * sqlite:///lab16.db
Done.
Done.
Done.
[]
```

```
import pandas as pd
df = %%sql --dataframe SELECT Doctors.name, Patients.name, appointment_date FROM Appointments JOIN Doctors ON Doctors.doctor_id = Appointments.doctor_id JOIN Patients ON Patients.patient_id = Appointments.patient_id WHERE Doctors.doctor_id = 1;
display(df)

 * sqlite:///lab16.db
None
```

```
%%sql
INSERT INTO Doctors (doctor_id, name, specialization) VALUES (1, 'Dr. Smith', 'Cardiology');

 * sqlite:///lab16.db
(sqlite3.IntegrityError) UNIQUE constraint failed: Doctors.doctor_id
[SQL: INSERT INTO Doctors (doctor_id, name, specialization) VALUES (1, 'Dr. Smith', 'Cardiology');]
(Background on this error at: https://sqlalche.me/e/20/gkpj)
```

```
import pandas as pd
df = %%sql --dataframe SELECT Patients.name, appointment_date, Doctors.name FROM Appointments JOIN Patients ON Patients.patient_id = Appointments.patient_id JOIN Doctors ON Doctors.doctor_id = Appointments.doctor_id WHERE Patients.patient_id = 1;
display(df)

 * sqlite:///lab16.db
None
```

```
import pandas as pd
df = %%sql --dataframe SELECT Doctors.name, COUNT(Patients.patient_id) AS total_patients FROM Appointments JOIN Doctors ON Doctors.doctor_id = Appointments.doctor_id JOIN Patients ON Patients.patient_id = Appointments.patient_id GROUP BY Doctors.name;
display(df)

 * sqlite:///lab16.db
None
```

```
import pandas as pd
df_doctors = %%sql --dataframe SELECT * FROM Doctors;
display(df_doctors)

 * sqlite:///lab16.db
None
```

Sample Data

Doctors

doctor_id	name	specialization
1	Dr. Sharma	Cardiology
2	Dr. Rao	Neurology

Patients

patient_id	name	age
1	Arjun	30
2	Meera	25

Appointments

appointment_id	doctor_id	patient_id	appointment_date
1	1	1	2026-03-10
2	1	2	2026-03-11

Task 3 – Library Database

AI Prompt

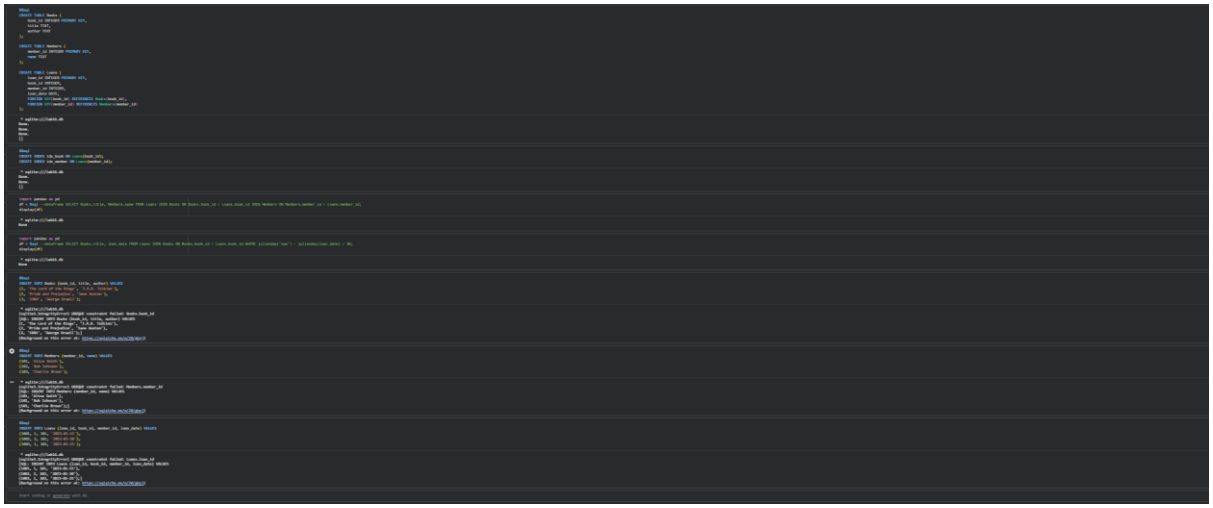
Generate SQL schema for a Library Management System.

Tables:
Books
Members
Loans

Requirements:

- Add indexing strategy
- Generate queries:
  Books currently issued
  Overdue books (loan > 30 days)
  Books loaned per member

CODE and OUTPUT:



Books

book_id	title	author
1	AI Basics	John
2	SQL Mastery	David

Members

member_id	name
1	Akshay
2	Ravi

Loans

loan_id	book_id	member_id	loan_date
1	1	1	2026-02-01

Task 4 – Online Shopping Database

AI Prompt

Generate SQL schema for an E-commerce platform.

Tables:

Users

Products

Orders

OrderDetails

Queries:

Retrieve all orders by a user

Find most purchased product

Calculate monthly revenue

Also suggest normalization improvements.

CODE and OUTPUT:

```
%%sql
CREATE TABLE Users (
    user_id INTEGER PRIMARY KEY,
    name TEXT,
    email TEXT UNIQUE
);

CREATE TABLE Products (
    product_id INTEGER PRIMARY KEY,
    name TEXT,
    price REAL
);

CREATE TABLE Orders (
    order_id INTEGER PRIMARY KEY,
    user_id INTEGER,
    order_date DATE,
    FOREIGN KEY(user_id) REFERENCES Users(user_id)
);

CREATE TABLE OrderDetails (
    order_detail_id INTEGER PRIMARY KEY,
    order_id INTEGER,
    product_id INTEGER,
    quantity INTEGER,
    FOREIGN KEY(order_id) REFERENCES Orders(order_id),
    FOREIGN KEY(product_id) REFERENCES Products(product_id)
);
```

```
 * sqlite:///ia6b6.db
Done.
Done.
Done.
Done.
{}
```

```
import pandas as pd
df = %%sql --dataframe SELECT Users.name, Orders.order_id, order_date FROM Orders JOIN Users ON Users.user_id = Orders.user_id WHERE Users.user_id = 1;
display(df)
```

```
 * sqlite:///ia6b6.db
None
```

```
import pandas as pd
df = %%sql --dataframe SELECT Products.name, SUM(quantity) AS total_sold FROM OrderDetails JOIN Products ON Products.product_id = OrderDetails.product_id GROUP BY Products.name ORDER BY total_sold DESC LIMIT 1;
display(df)
```

```
 * sqlite:///ia6b6.db
None
```

Gemini

```
import pandas as pd
df = %%sql --dataframe SELECT strftime('%Y-%m', order_date) AS month, SUM(Products.price * OrderDetails.quantity) AS revenue FROM Orders JOIN OrderDetails ON Orders.order_id = OrderDetails.order_id JOIN Products ON Products.product_id = OrderDetails.product_id GROUP BY month;
display(df)
```

```
 * sqlite:///ia6b6.db
None
```

Users

user_id	name
1	Akshay

Products

product_id	name	price
1	Laptop	60000
2	Mouse	500

Orders

order_id	user_id	order_date
1	1	2026-03-01

OrderDetails

order_detail_id	order_id	product_id	quantity
1	1	1	1
2	1	2	2

Task 5 – University Examination Database

AI Prompt:

Generate SQL schema for a University Examination System.

Tables:

Students

Subjects

Exams

Results

Queries:

Insert exam results

Retrieve subject-wise marks

Calculate average marks per subject

CODE and OUTPUT:

```
%%sql
CREATE TABLE Subjects (
    subject_id INTEGER PRIMARY KEY,
    subject_name TEXT
);

CREATE TABLE Exams (
    exam_id INTEGER PRIMARY KEY,
    subject_id INTEGER,
    exam_date DATE,
    FOREIGN KEY(subject_id) REFERENCES Subjects(subject_id)
);

CREATE TABLE Results (
    result_id INTEGER PRIMARY KEY,
    student_id INTEGER,
    exam_id INTEGER,
    marks INTEGER,
    FOREIGN KEY(student_id) REFERENCES Students(student_id),
    FOREIGN KEY(exam_id) REFERENCES Exams(exam_id)
);

 * sqlite:///ludb0.db
Done.
Done.
Done.
[]
```

```
%%sql
INSERT INTO Results (result_id, student_id, exam_id, marks)
VALUES (1,1,1,85);

 * sqlite:///ludb0.db
1 rows affected.
[]
```

```
import pandas as pd
df = %sql --dataframe SELECT Students.name, Subjects.subject_name, Marks FROM Results JOIN Students ON Students.student_id = Results.student_id JOIN Exams ON Exams.exam_id = Results.exam_id JOIN Subjects ON Subjects.subject_id = Exams.subject_id;
display(df)
```

```
 * sqlite:///ludb0.db
None
```

```
import pandas as pd
df = %sql --dataframe SELECT Subjects.subject_name, AVG(marks) AS average_marks FROM Results JOIN Exams ON Exams.exam_id = Results.exam_id JOIN Subjects ON Subjects.subject_id = Exams.subject_id GROUP BY Subjects.subject_name;
display(df)
```

```
 * sqlite:///ludb0.db
None
```

Subjects

subject_id	subject_name
1	Database
2	AI

Results

result_id	student_id	exam_id	marks
1	1	1	85
2	1	2	90

Query – Subject Wise Marks

Output

student	subject	marks
Akshay	Database	85
Akshay	AI	90

Final Analysis

The experiment demonstrated how AI can assist in designing relational database schemas and generating SQL queries. Multiple database systems were modeled including Student Information Systems, Hospital Management Systems, Library Systems, E-commerce platforms, and University Examination Systems. SQL queries demonstrated CRUD operations, joins, and aggregate functions. Normalization principles and indexing strategies were applied to improve data integrity and query performance.